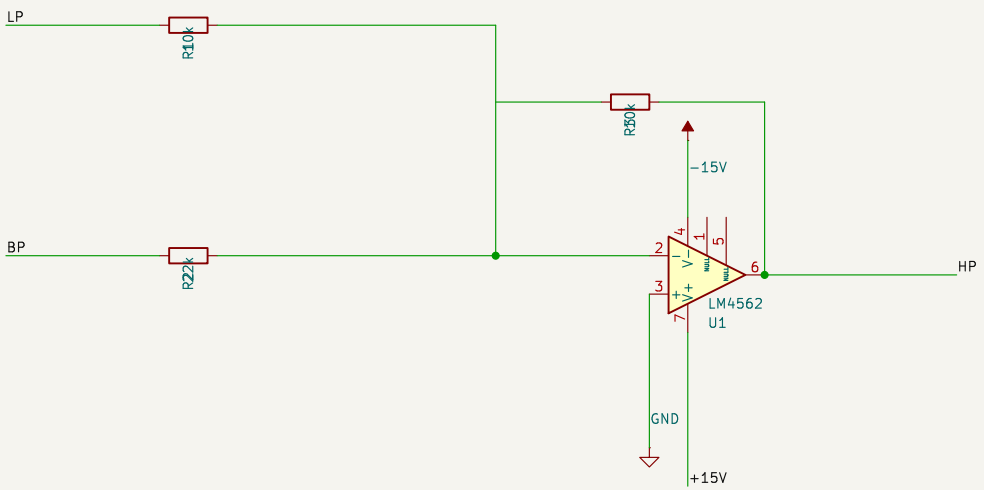
NG AMPLIFIER (U1)

R1/R3) * LP − (R2/R3) * BP

0k = −1.0 BP gain = −22k/10k = −2.2



Block 1 of 6
LP gain = −1.0, BP gain = −2.2
HP = −(R1/R3)*LP − (R2/R3)*BP
CircuitForge

Sheet: /
File: osc_block_summing_amp.kicad_sch

Title: State Variable Oscillator − Summing Amplifier

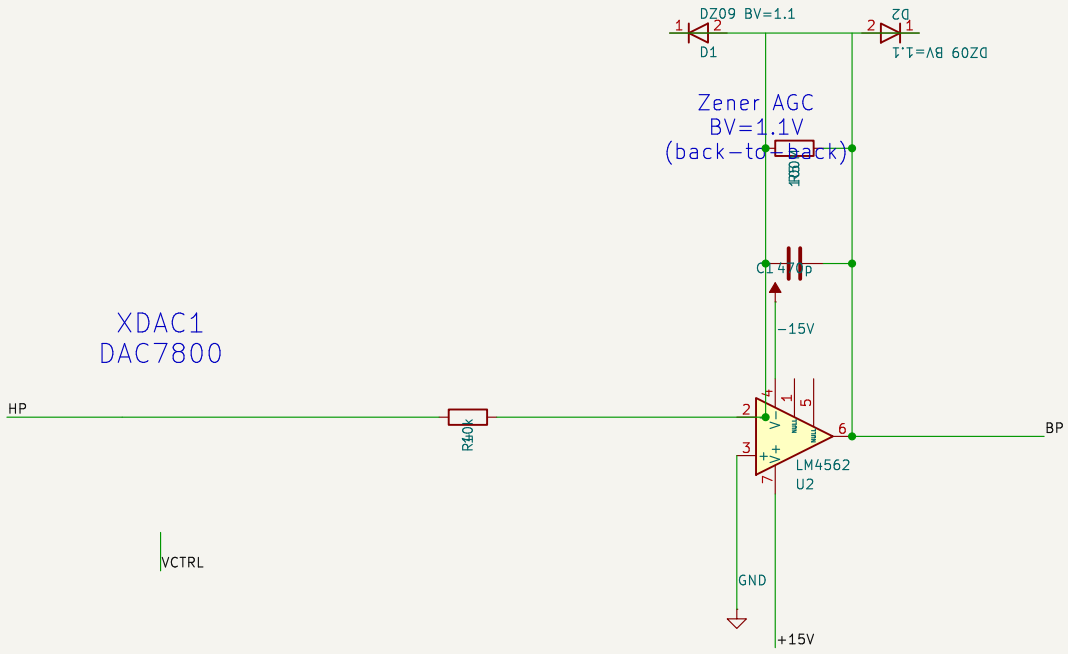
Size: A4	Date:	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 1/1

Interfaces:

: HP -> BP (U2 + XDAC1)

R4 * C1) = D / (4096 * 2pi * 10k * 470p)

nge: 25 Hz (D=3) to 30 kHz (D=3640)



Block 2 of 6
Zener AGC: BV=1.1V, BP amplitude ~1V RMS
f = D/(4096*2pi*10k*470p), 25Hz–30kHz
CircuitForge

Sheet: /
File: osc_block_integrator1.kicad_sch

Title: State Variable Oscillator – Integrator 1 (HP to BP)

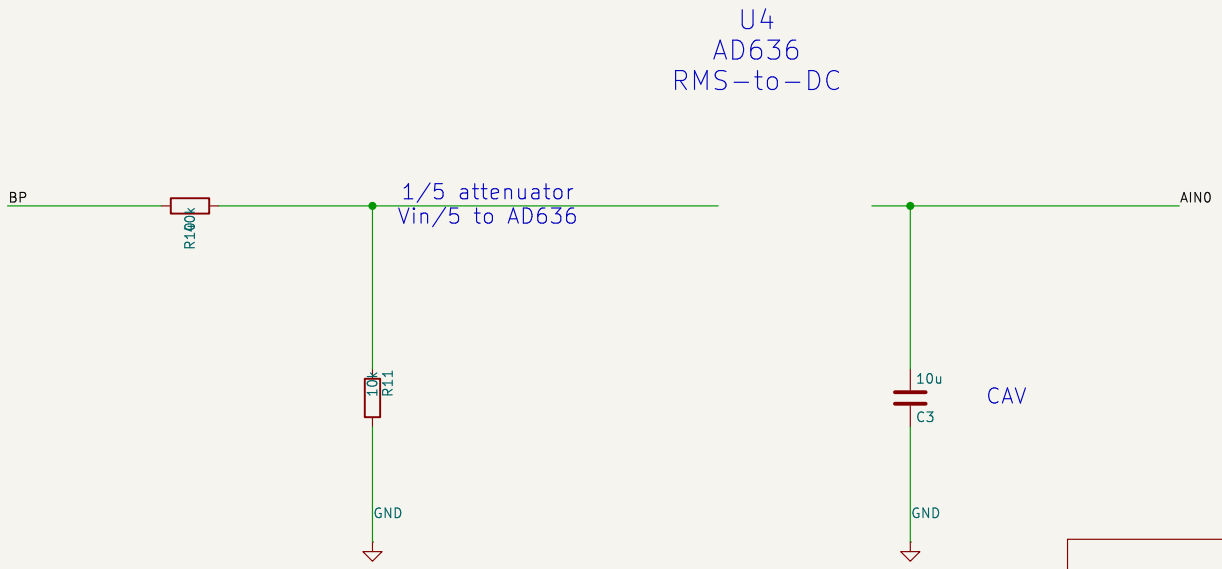
Size: A4	Date:	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 1/1

Hz D=3: f~25Hz D=3640: f~30kHz

RMS DETECTOR (U4)

(R10+R11) = BP * 10k/(40k+10k) = BP/5

-DC converter, CAV=10uF averaging capacitor



Block 3 of 6
Vout = BP * R11/(R10+R11) = BP/5
1/5 attenuator + AD636 true RMS-to-DC
CircuitForge

Sheet: /
File: osc_block_rms_detector.kicad_sch

Title: State Variable Oscillator - AD636 RMS Detector

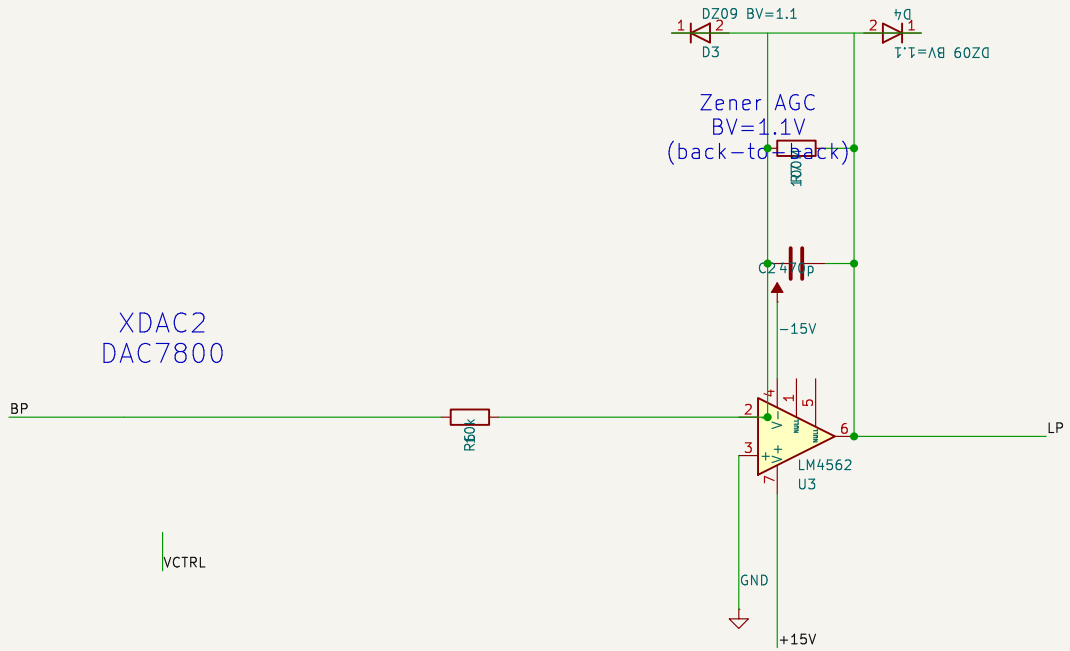
Size: A4	Date:	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 1/1

Interfaces:

: BP -> LP (U3 + XDAC2)

R6 * C2) = D / (4096 * 2pi * 10k * 470p)

Integrator 1, R8=100k output load



Block 4 of 6
Zener AGC: BV=1.1V, R8=100k output load
f = D/(4096*2pi*10k*470p), mirrors Integrator 1

CircuitForge

Sheet: /
File: osc_block_integrator2.kicad_sch

Title: State Variable Oscillator – Integrator 2 (BP to LP)

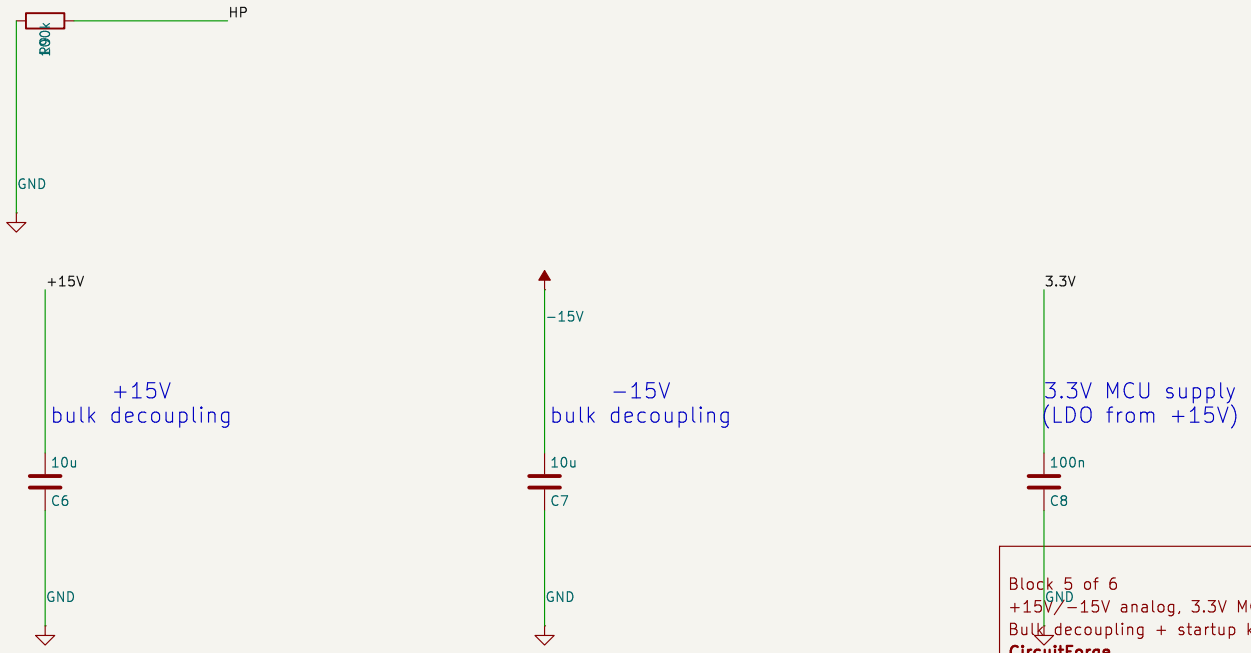
Size: A4	Date:	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 1/1

KICK + POWER SUPPLY

1V pulse into HP net to initiate oscillation

analog supply, 3.3V MCU via LDO from +15V

Startup Kick
(0, 0.1V, 0.1ms, 1ns, 1ns, 10us)



Block 5 of 6
+15V/−15V analog, 3.3V MCU (LDO from +15V)
Bulk decoupling + startup kick
CircuitForge

Sheet: /
File: osc_block_power_supply.kicad_sch

Title: State Variable Oscillator – Power Supply + Startup

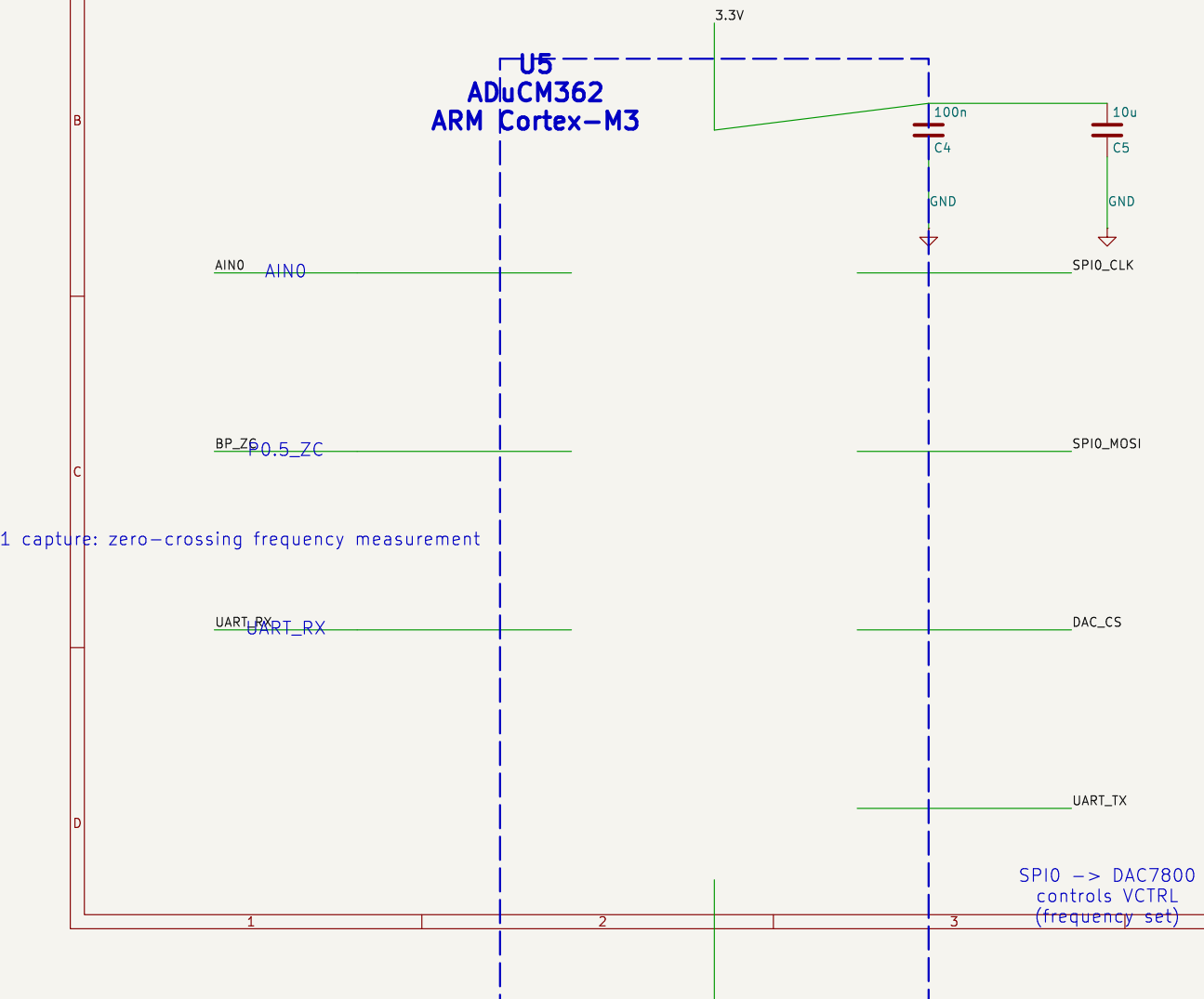
Size: A4	Date:	**Rev: 1.0**
KiCad E.D.A. 9.0.7		Id: 1/1

Interfaces:

CM362 MCU (U5)

24-bit Sigma-Delta ADC, SPI, UART

AC7800 MDACs, Timer1 measures BP frequency



Block 6 of 6
Timer1 zero-crossing, UART 115200 8N1
SPI0->DAC7800 freq ctrl, ADC AIN0<-AD636
CircuitForge
Sheet: /
File: osc_block_mcu.kicad_sch

Title: State Variable Oscillator - ADuCM362 MCU

Size: A4	Date:	Rev: 1.0
KiCad E.D.A. 9.0.7		Id: 1/1